

There are various ways in which traders and risk managers might approach this issue. But all of these methods require that the instruments in the portfolio are modeled in a way so that the model correlations show a degree of horizon-dependence consistent with the empirical data. And for that, we find the MVOU process to be a particularly useful model.

## EUR 5Y5Y AND GBP 5Y5Y IMPLIED VOLATILITIES

As another example, let's revisit the analysis of EUR 5Y5Y and GBP 5Y5Y implied swaption volatilities from Chapter 2.

Figure 4.14<sup>5</sup> shows these two series over time, from 25-May-05 through 3-Jul-12.

We have *a priori* economic reasons for modeling these implied vols as stationary series. And it does appear that the two volatilities are pretty highly correlated with one another.

The difference between the GBP series and the EUR series over time is shown in Figure 4.15.

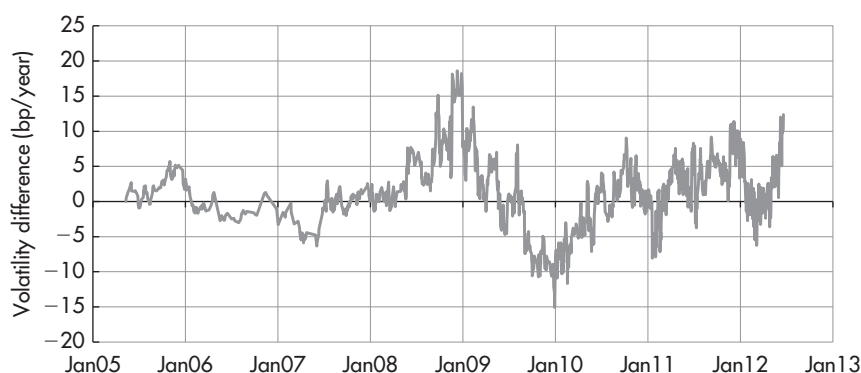
From this graph, it certainly appears that the spread is mean-reverting. Of course, the difference between two stationary series is also stationary, so this observation shouldn't come as a surprise. The more interesting issue is the extent to which the positive correlation evidenced in Figure 4.14 is the result of any attraction that might exist between the variables, of the sort inherent in the MVOU model.



**FIGURE 4.14** EUR and GBP 5Y5Y implied swaption volatilities.

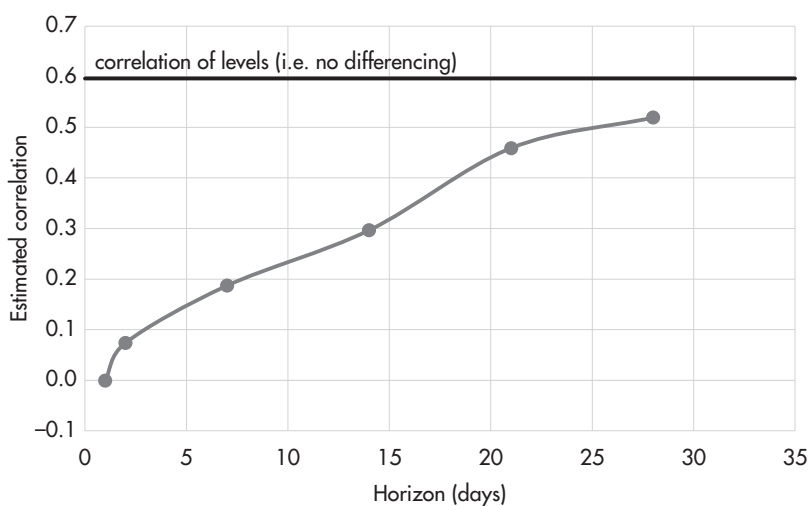
Source: data – Bloomberg, chart – Authors.

<sup>5</sup>Figures 4.14 and 4.15 are identical to Figures 2.17 and 2.18 and are repeated here for convenience.



**FIGURE 4.15** Swaption volatility difference: EUR 5Y5Y – GBP 5Y5Y.

Source: data – Bloomberg, chart – Authors.



**FIGURE 4.16** Estimated correlation coefficient as a function of horizon.

Source: Authors.

Figure 4.16 shows the estimated correlation coefficient between the two series as a function of horizon (i.e. the time between successive, non-overlapping data used in the calculation).

Consistent with the discussion above, the estimated correlation coefficient is close to zero on a horizon of one day and becomes increasingly positive as the horizon increases. By a horizon of 28 days, the estimated correlation

is 0.52, not far from the correlation of 0.60 calculated using the original, undifferenced data.

At this point, particularly given horizon-dependent correlation, the MVOU process appears to be a good choice for modeling these two series.

Fitting an MVOU model to this data produces an estimated transition matrix of  $\Theta = \begin{bmatrix} 8.93 & -8.06 \\ -7.42 & 8.66 \end{bmatrix}$ . The first element in the top row shows the strength of mean reversion in the EUR series. A figure of 8.93 would correspond to a half-life of only 28 days, if there were no other influences on the EUR series. The corresponding figure for the GBP series is shown as the second element in the second row of this matrix.

The second element of the first row shows the extent to which the EUR series appears to be attracted to the GBP series. Recall that a negative number means that the EUR series is attracted to the GBP series. (A positive number would suggest that the EUR series is repelled by the GBP series.)

The first element of the second row of this matrix shows the extent to which the GBP series is attracted to the EUR series. Again, this estimate suggests that the GBP series is attracted to the EUR series.

We can gain some further intuition about these dynamics by multiplying the transition matrix by some time horizon, such as one day, and then calculating the exponential of the resulting matrix. In this case, the result is  $e^{-\Theta(t-s)} = \begin{bmatrix} 0.976 & 0.020 \\ 0.022 & 0.977 \end{bmatrix}$ . We can gain some further intuition by writing the two equations that this matrix represents.

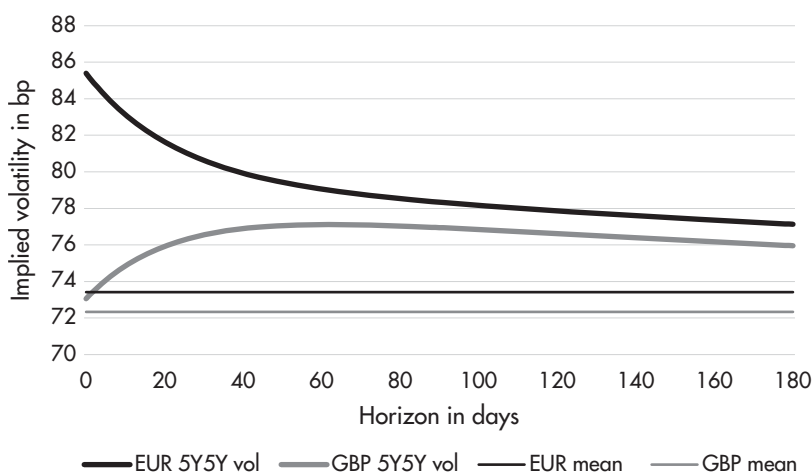
$$E[EUR_t] = \mu_{EUR} + 0.976(EUR_s - \mu_{EUR}) + 0.020(GBP_s - \mu_{GBP})$$

$$E[GBP_t] = \mu_{GBP} + 0.022(EUR_s - \mu_{EUR}) + 0.977(GBP_s - \mu_{GBP})$$

for  $t - s = \text{one day}$ , where  $\mu_{EUR}$  and  $\mu_{GBP}$  are the long-run means of the EUR and GBP implied vols.

In words, the EUR series is expected to close 2.4% of the gap between its current value and its long-run mean but is also expected to move a bit higher when the GBP series is elevated relative to its long-run mean. The figures for the GBP series are quite similar.

The estimated scatter matrix is  $\begin{bmatrix} 21.48 & -0.043 \\ 0.106 & 18.45 \end{bmatrix}$ . If we convert that to a covariance matrix over a horizon of one day, we get  $\begin{bmatrix} 1.23 & 0.026 \\ 0.026 & 0.910 \end{bmatrix}$ . The corresponding standard deviation for the EUR series is 1.11 bp/year, while the standard deviation for the GBP series is 0.95 bp/year. The correlation coefficient obtained from this matrix is 0.025.



**FIGURE 4.17** Expected values as a function of time.

Source: Authors.

These figures are consistent with the estimated standard deviations and correlation coefficient obtained using daily observations, particularly given that a fair number of the ‘daily’ observations in our sample are separated by more than one day, owing to the existence of weekends and holidays.

The paths of the expected values for each series are shown in Figure 4.17.

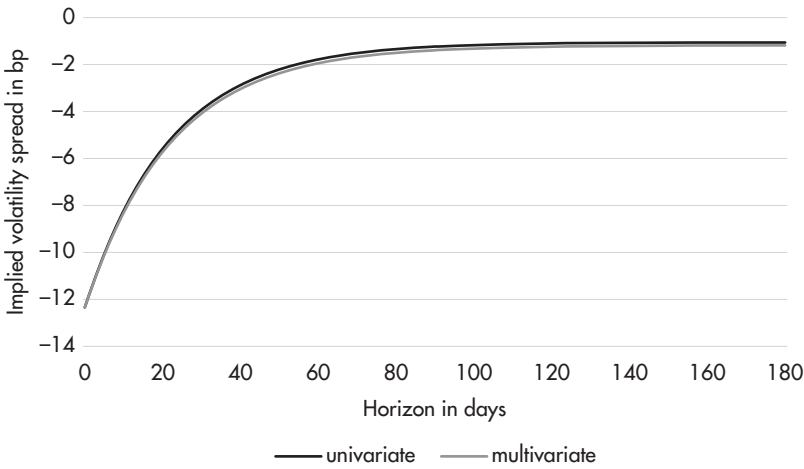
Note that both series start above their long-run means. But while the EUR series initially is expected to decline toward its mean, consistent with our intuition regarding mean-reverting series, the GBP series is expected to *increase* toward the EUR series initially before beginning a decline.

The attraction that seems to exist between the two variables is responsible for the fact that two variables that appear to be uncorrelated over very short horizons can appear highly correlated over longer horizons. The effect of the attraction cumulates over time, with the result that the correlation over longer horizons is quite a bit greater than the correlation over very short horizons.

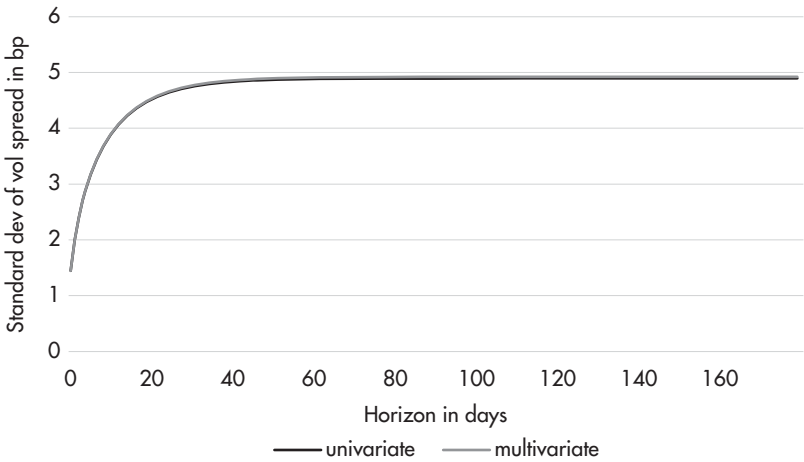
With all this in mind, was the univariate OU process used in Chapter 2 a bad choice? It depends. Figure 4.18 shows the path of expected values for the spread using the MVOU model and using the OU model. The two appear virtually indistinguishable.

Figure 4.19 shows the path of the conditional standard deviation of the spread, as calculated by the two models.

The two models appear to produce nearly identical results. As it happens, this is consistent with a basic result that a linear combination of variables that follow a MVOU process can be modeled as a univariate OU process if



**FIGURE 4.18** Expected spread as a function of time.  
*Source:* Authors.



**FIGURE 4.19** Standard deviation of the spread as a function of time.  
*Source:* Authors.

and only if the linear combination is orthogonal to one of the eigenvectors of the transition matrix. The eigenvectors of the transition matrix discussed above appear as the columns of the matrix  $\begin{bmatrix} -0.728 & -0.716 \\ -0.686 & -0.699 \end{bmatrix}$ , and the second of these is essentially orthogonal to the weight vector used to construct this simple spread,  $[-1 \ 1]$ . So this happens to be a case in which our spread can be modeled with a univariate process.

Given these results, is the MVOU model superfluous in this case? Not at all. It provided insights about the correlation structure of the data as a function of the horizon that the univariate model can't provide. And it showed that the expected path of the GBP implied volatility series is nonmonotonic, likely to increase initially before decreasing later – again, something the univariate model is unable to address.

## CONCLUSIONS AND IMPLICATIONS

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Are the additional benefits of the MVOU model worth the additional costs required to implement the model relative to the univariate OU model and principal components analysis?

In our view, the cost–benefit analysis will favor the MVOU in some cases but not in all cases. For example, if it's useful to know whether one or more of the variables is likely to exhibit a nonmonotonic path of expected values, the MVOU process is probably worth the additional effort. Similarly, if a quick check of the correlation structure of the available data suggests horizon-dependent correlation, we would suggest using the MVOU model.

On the other hand, if the empirical data doesn't exhibit horizon-dependence, it may be more straightforward to use PCA to model the data. And if the weights used to construct a spread (or portfolio) are orthogonal to one of the eigenvectors of the transition matrix, then the linear combination can be modeled with a univariate OU process anyway, though – as we saw in one of our examples – it might still be useful to know the specific paths of expected values in any event.

MVOU models can be implemented in packaged software so that the computational costs are hidden from the analyst. However, there are other costs to implementing the MVOU process that are worth considering. For example, the MVOU process is the continuous-time limit of a first-order vector autoregressive (VAR) process. In general, econometricians approach vector autoregressions with a bit of skepticism, as the greater number of parameters affords the analyst more of an opportunity to overfit the data. In fact, one of our former professors, Arnold Zellner, used to joke that VAR stood for “very awful regressions” precisely for this reason.

A cottage industry has arisen to help analysts avoid the pitfalls of overfitting, and we encourage analysts using MVOU models to familiarize themselves with some of these methods. In our own experience, we've found it useful to impose *a priori* constraints on some of the parameters, in order to reduce the effective degrees of freedom that otherwise might lead us to overfit data. For example, when modeling multiple points on a yield curve, we sometimes impose the constraint that only adjacent points on the curve can influence one another. This assumption alone will reduce the number of free parameters in the transition matrix from  $m^2$  to  $3m - 2$ .

Another *a priori* restriction we sometimes impose is that any influence between variables is symmetric. In other words, the influence exerted by one variable on another is identical to the influence that the second variable exerts on the first. When this constraint is justified by the economics of the situation, it will reduce the number of free parameters in the transition matrix to  $\frac{m^2+m}{2}$ .

Of course, there are risks in imposing *a priori* constraints, even when they're motivated by sound economic logic. But we're persuaded that there are times when, as per Judea Pearl's maxim: "*you are smarter than your data.*" That's true when imposing constraints on parameters within a specific model like the MVOU model. And it's also true when choosing among models. Over time, we've become more comfortable using our experience to choose among the three statistical models discussed in this book. And we expect the same will be true of other analysts as they gain practical experience implementing these models.

# Financial Models





# Some Comments on Yield, Duration, and Convexity

## INTRODUCTION

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This book is intended for somewhat experienced fixed income analysts, traders, and portfolio managers, so we assume the reader has more than a passing familiarity with the concepts of yield, duration, and convexity. However, we do come across some confusion concerning these concepts from time to time, so we'll offer a few thoughts in this chapter, designed to help clarify potential misunderstandings.

## SOME BRIEF COMMENTS ON THE YIELD OF A COUPON-PAYING BOND

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A bond that pays a coupon is simply a portfolio of zero-coupon bonds, and the price of the coupon bond is the sum of the prices of the zero-coupon bonds.

However, the yield of a coupon bond generally can't be expressed as a linear combination of the yields of the constituent zero-coupon bonds. Nevertheless, many investors would like to be able to talk about the yield or the rate of return of a coupon bond. To that end, the yield of a coupon bond is typically defined as the discount rate that when applied to all cash flows produces a net present value equal to the invoice price of the bond. In other words, it's the value of  $y$  that satisfies

$$P = \sum_{i=1}^K \frac{X_i}{\left(1 + \frac{y}{c}\right)^{cT_i}}$$

where  $P$  is the price of the bond;  $K$  is the number of cash flows,  $X_i$  is the size of cash flow  $i$ ,  $c$  = the compounding frequency, and  $T_i$  is the time until the  $i$ -th cash flow is due to be received.